Dataset engineering, biological debrief, target selection, and co-scientist architecture

Prepared 18 October 2026; §2.5 and the summary added 19 October. Feeds gates COSCI-07 (two prototype families, opens Oct 19), LAB-01 (single-round experimental question, Oct 22–23) and GATE-06 (feasibility and novelty review, Oct 23).

No claim of therapeutic efficacy, antiviral activity, binding affinity or full-system performance is made anywhere below. Targets in §3 are proposals for review with Askar and the SynX experimental lead, not selections.

Reading this document. The summary that follows is self-contained. §3, on targets, is ordinary target-selection argument and assumes no computational background. The remaining sections use machine-learning and statistics vocabulary, each term of which is defined in the glossary at §6; those sections open with a paragraph setting out what they establish before the detail begins.

Summary for review

Scope of the October work. Two small computational pilots were conducted. Neither concerns whether a molecule binds anything. Both address a narrower question: under a limited measurement budget and delayed results, does the *rule used to select what to measure next* materially affect the outcome? Both were run against published measurement datasets in which the result for every candidate is already known, so the selections can be graded automatically and no new laboratory work was required.

First finding: a measurement-design result, which replicates. Grading a decision system on everything it *selected* overstates its performance relative to grading it on what had actually returned at the time each decision was made. The discrepancy is a factor of approximately 1.2 to 1.4. The distinction is not bookkeeping. It separates the question "were these good selections in hindsight?" from "were these good selections on the evidence then available?" Only the latter is a fair test of an emergency decision-maker, and the former is what most evaluations measure inadvertently. The effect reproduced on every dataset examined, including SARS-CoV-2 Mpro and both influenza haemagglutinin datasets.

Second finding: an earlier headline result does not survive scrutiny. A previous figure reported that the selection rule recovered approximately eight times more of the best candidates than random selection. That figure is substantially an artefact of pool size. At 128 measurements drawn from 4,096 candidates, random selection recovers about 3% of the strong candidates, leaving considerable scope for apparent improvement. At 128 drawn from 512, random selection already recovers about 25%, and no method can exceed it by more than fourfold even in principle. Once corrected, the advantage on pathogen datasets is small and inconsistent — clearest on influenza H3, absent on H1 — and one of the four rules performs worse than random. **The eightfold figure should not appear in any paper or presentation.** The supporting arithmetic is given in §2.5.

System capability as of 19 October. Until that date the software selected among three abstract strategies while representing candidates only as numbered items; it held no representation of the biology involved. It now operates on a target dossier structured around the seven criteria in §3.2 — reagent tractability, essentiality, conservation, accessibility, assay availability, public data and precedent. It records which claim rests on which source, refuses evidence dated later than the decision under consideration, retains contradictory evidence alongside the claim it contests rather than overwriting it, and states what a single selection round can and cannot establish. Applied to the coronavirus packet it proposes **Mpro**, notes that a cytosolic target requires a membrane-permeability argument, limits the claim available from one round to enrichment, and flags that its own assertion regarding prior macrocycle campaigns against Mpro is uncited and requires verification.

Decisions required this week.

1. **Two prototype families.** The proposal is Coronaviridae via Mpro together with Orthomyxoviridae via the haemagglutinin stem or neuraminidase, with Nipah retained as a stretch option. The reasoning is set out in §3.3 and §3.4. The deciding consideration is the availability of public measurement data, which allows method development to proceed without dependency on SynX; Nipah offers the stronger narrative and no such data.
2. **The question for the single display round** (§3.8). The proposal is Mpro: the least expensive well-behaved reagent, the strongest conservation case, and an orthogonal functional assay available if scheduling permits.
3. **Three dependencies outside this project's control**, each already past its date: a named experimental owner with a booked slot; the campaign export, without which the platform-improvement deliverable cannot begin; and confirmation of trainable model access with an approved spending ceiling.

Claims not made in this document. No binding affinity, functional inhibition, antiviral activity or therapeutic efficacy is asserted. No macrocyclic peptide and no non-canonical amino acid has been measured or modelled; the building-block work to date is a catalogue structure awaiting chemistry from the platform. The pilots measure decision procedure, and a measurement of procedure is not evidence about a molecule.

1. How the dataset was engineered

Two decision-policy datasets were constructed from public sources. The engineering value is almost entirely in the **leakage controls and the accounting**, not in the substrates, so that is what this section documents.

1.1 Binding dataset

What follows describes a simulated experiment campaign. A published dataset in which every variant's result is already known is taken as the substrate; all results are withheld; the system requests them a few at a time under a fixed budget; and its selections are then graded against the withheld values. Almost all of the detail below exists to guarantee a single property: that the system cannot reach an answer it has not earned.

Source. ProteinGym v1.3, assay `SPG1_STRSG_Wu_2016` — 149,360 measured variants. ProteinGym records `selection_assay: Binding (IgG)`, `selection_type: binding`, so the readout is affinity selection rather than a growth or fitness proxy.

Candidate representation. Only the four variable residue identities are exposed. Features are one-hot encoded per position and flattened — `np.eye(20)[features]` reshaped to 80 dimensions per candidate. No sequence context, no structural features, no evolutionary features. This is deliberate: it keeps the surrogate weak and interpretable so that measured gains are attributable to the acquisition policy rather than to feature engineering.

Pool construction. Twenty evaluation pools of 4,096 candidates plus one separate 4,096-candidate development pool. All pools are mutually disjoint and were selected **without outcome filtering** — no pool was chosen or rejected on the basis of the scores it contains. Pool-to-pool variability is therefore descriptive spread across sub-landscapes; it is not biological replication across independent proteins, and must never be reported as such.

Hidden-outcome isolation. The full outcome vector stays in the evaluator. The policy interface accepts only revealed values, enforced at the function boundary in `pick_batch` and covered by a test. Top-1% membership is never supplied to a policy. Candidate identities in LLM prompts are opaque integer pool indices, so the model cannot recognise a variant and recall its phenotype.

Paired design. The eight initial measurements are identical across all policies within a pool. Every policy comparison inside a pool therefore starts from the same evidence, which removes initialisation variance from the comparison.

Acquisition loop. Total budget 128 measurements, acquired in batches of 8, with checkpoints recorded at 8 / 16 / 32 / 64 / 128. Repeated or previously-queried acquisitions raise rather than being silently deduplicated.

Surrogate. A single fixed 64-tree random forest (`min_samples_leaf=2`, `max_features=1.0`) is shared by greedy, UCB and the LLM-mediated policy, so no policy gets a better model than another. Per-tree prediction dispersion is used as the uncertainty heuristic. It is a heuristic, not a calibrated posterior — see §2.3.

Tie handling. Candidates are permuted with a seeded RNG before a stable argsort, so ties at the acquisition threshold are broken reproducibly without perturbing the score ordering. This matters more than it sounds: the GB1 landscape has large exactly-tied plateaus at low budget, and unseeded tie-breaking silently injects a diversity bonus into the greedy policy.

Delayed feedback. Acquisitions enter a pending queue keyed by `(round + delay, ids)` and become visible only when the round arrives. Conditions are 0 and 3 **decision rounds**. These are synthetic stress tests and are explicitly not estimates of assay turnaround.

Dual evaluation. Every checkpoint records both the retrospective metrics (`best_score`, `top_recall` over everything *selected*) and the realtime metrics (`returned_best_score`, `returned_top_recall`, `pending_labels` over what has actually *returned*). Keeping both is the single most useful design decision in the dataset — see §2.2.

1.2 Structural dataset

Source. ATLAS, pinned metadata `2024_11_18_ATLAS_info.tsv`, three 100 ns MD replicates per entry.

Selection. Human organism, 50–250 residues, nonredundant flag, one entry per ECOD family, sorted by PDB identifier, split 4 development / 8 evaluation. Critically, **trajectory-derived outcome columns were excluded from the selection criteria**, so entries were not chosen for having convenient dynamics.

Observable. Solvent-accessible surface area via MDTraj `shrake_rupley`, nm² per residue, 240 sphere points, trajectory stride 5, in acquisition blocks of 20 saved frames.

Evidence isolation. Each protein's first two blocks expose one block from replicate R1 and one from R2; later requests interleave chronological blocks from R1 and R2. **R3 is held out entirely** and is used only inside the evaluator as a simulation reference. Temporal correlation within a replicate is preserved rather than shuffled away, so block dispersion is a heuristic and not a frame-independent confidence interval.

Cost accounting. Acquisition cost is replayed from *measured* per-block preprocessing CPU time, recorded during preprocessing. It is not live MD cost and not request latency. LLM latency and tokens are logged separately so compute cost and inference cost never pool.

1.3 Provenance and reproducibility

- SHA-256 hashes for every downloaded input and every frozen split, in `data/processed/*/manifest.json`.
- Every result file embeds `dataset_manifest_sha256` and `config_sha256`, so a result can be tied to the exact config and data that produced it.

- Atomic JSON writes (temp file then replace) with `allow_nan=False`, so a NaN cannot silently enter a results file and a partial write cannot be mistaken for a result.
- Smoke outputs carry a `smoke` flag and are excluded from scientific summaries.
- Inference or analysis failure fails its task explicitly. A failed LLM call never degrades into a baseline policy result.

1.4 Agent interface

The LLM policy speaks to any OpenAI-compatible backend selected by `PILOT_LLM_PROVIDER` (local vLLM, OpenRouter, Anthropic, OpenAI), with per-provider request shaping. Actions are constrained by a strict JSON schema; an action outside the allowed set, or a reply containing no JSON object, raises. Hosted runs get a provider-specific policy label so they can never pool with or overwrite local-model runs, and the completion accounting counts only the configured grid.

1.5 What this design does and does not model

This bears statement, because it governs §2.

What the dataset models is **iterative small-batch assay follow-up** over a dense, fully-enumerated candidate space — four sites by twenty canonical amino acids — where every candidate carries a quantitative score and feedback arrives after a synthetic delay measured in decision rounds.

What it does not model is a **single large display selection round**: a sparse library of order 10^{12} macrocycles, read out as enrichment ratios across successive rounds rather than as per-candidate scores, built from non-canonical residues and a cyclisation chemistry the encoding has no representation for, and returned on real reagent and assay turnaround rather than a round counter.

The gap is not a detail of scale. Nothing in the pilot exercises the two features that define the SynX platform — non-canonical chemistry and enrichment-based readout — so no result from it can speak to either.

2. Biological debrief, and whether the earlier experiments survive

2.1 What the substrates actually are

Binding arm. `SPG1_STRSG_Wu_2016` is the **B1 domain of streptococcal protein G (GB1)** — residues 265–280 of a 448-residue parent, four co-varying sites, measured by IgG Fc affinity selection. A bacterial immunoglobulin-binding domain with canonical amino acids at four positions. Not a peptide, not cyclic, no non-canonical residues, no pathogen relevance.

Structural arm. All twelve ATLAS entries are **human proteins**, selected for ECOD family diversity with no therapeutic intent.

- *Evaluation (8):* TFIIF subunit 2, platelet basic protein (CXCL7), retinoic acid receptor γ, SKP2, non-secretory ribonuclease, IGF-1, UBR5, major prion protein.
- *Development (4):* galectin-3-binding protein, Axin-1, moesin, MEF2A.

No macrocyclic peptide, no non-canonical amino acid and no pathogen target appears anywhere in the codebase or results.

2.2 The argument for keeping the binding arm

(a) It is a validated scoring harness, and the training plan requires one. The Week 8–10 harness work depends on baseline parity — a scorer cannot be validated against nothing. The existing tests cover disjoint pools, unique acquisitions, delayed-outcome visibility, policy isolation from hidden labels, structural reference isolation and resource accounting. That is reusable infrastructure regardless of substrate, and rebuilding it against a pathogen target is a config change plus pool-size rework, not a rewrite.

(b) The selected-versus-returned gap is a measurement-design finding, not a biology finding — and it is real. Scoring a policy on what it *selected* rather than what has *returned* overstates it by roughly 1.4× at a three-round delay (top-1% recall 0.204 vs 0.144 for greedy; 0.213 vs 0.162 for UCB). That is a statement about how to construct a decision metric under delayed feedback, and it transfers directly into the episode rubric and the uplift loss definition. Any evaluation of an emergency scientist that scores retrospective selection quality is measuring something the agent could not have known.

(c) The calibration result de-risks the project premise, and this is the strongest argument. Acquisition policy delivers roughly 8× top-1% recall over random *while the surrogate is only weakly informative* — Spearman ≈ 0.31 against truth at the 128-label budget, with plain ridge regression matching a 64-tree forest. If decision-policy gains had required a good predictor, the whole emergency premise would collapse, because at emergency timescales there are no labels to fit one. The result says the gains come from the policy. That is the claim worth carrying forward, and it is substrate-independent in a way the recall numbers are not.

What does not survive. The quantitative numbers transfer not at all. GB1 is dense, small and fully enumerated; display is sparse, vast and macrocyclic with an enrichment readout. "128 labels from a 4,096 pool in batches of 8" models iterative plate-based or biophysical follow-up, not a selection round. The 0-vs-3-round delay is unanchored to any real turnaround. None of this should appear in a paper as a finding about peptide discovery.

Verdict: keep, re-point, and demote. Treat the binding arm as harness validation plus hypothesis (b) and (c), re-run against a pathogen target, and never present the GB1 recall figures as a scientific result.

2.3 The argument against keeping the structural arm

No persuasive argument for retention could be constructed. The case for retirement follows.

The arm measures how fast mean per-residue SASA estimates converge to a held-out MD replicate, under different block-allocation policies, on twelve human proteins chosen for family diversity. The measured effect is real but small — uncertainty-weighted allocation reached the same accuracy as uniform at about 33% less replayed CPU (68 s vs 101 s), then plateaued and drifted slightly worse.

The problem is that **SASA agreement with a held-out MD replicate is not an epitope-accessibility measure** in any sense that feeds target ranking. Nothing in the theory of change consumes it. Surface exposure does not establish binding, inhibition or antiviral activity, and the README already says so. Re-pointing it at viral glycoproteins would not fix this, because the thing being measured — convergence of an average to a sibling simulation — still would not answer an epitope question.

Verdict: retire the arm as a scientific component. Keep the cost-accounting and evidence-isolation code, which is reusable. If structural evidence allocation is still wanted as a decision domain, it needs rebuilding around a question a dossier actually asks: *is this epitope solvent-exposed and conformationally stable enough to be worth a display campaign*, scored against something other than another MD replicate.

2.4 Calibration detail, for the record

Calibration asks whether a predictor's own confidence can be trusted — whether it is reliably wrong more often when it reports being unsure. The two tables below say that at small numbers of measurements this predictor is severely overconfident, understating its own error roughly sixtyfold, but that the order in which it ranks candidates becomes informative once about 64 measurements are in. Its ranking is usable; its error bars are not.

BUDGET	OVERCONFIDENCE RATIO	1Σ COVERAGE	SPEARMAN(Σ, ABS ERR)
8	63.5	0.56	0.21
32	30.2 (greedy)	0.73	0.56
128	1.8 (greedy) / 3.1 (random)	0.91 / 0.84	0.87

Forest-dispersion sigma is severely overconfident as a magnitude at low budget but ranks error reliably past roughly 64 labels. Usable for ordering, not as an interval.

BUDGET	SPEARMAN(PRED, TRUTH), FOREST	TOP-1% PRECISION, FOREST	SPEARMAN, RIDGE
8	0.05	0.026	0.13
128	0.31	0.088	0.28
1024	0.51	0.235	0.46

Reproducibility. Both studies are generated by `hpc_pilots/calibration.py` (`python -m hpc_pilots calibration`). `rf_calibration` reproduces the recorded results bit-exactly across all 200 rows and all eight columns, as does every forest row of `surrogate_skill`. The ridge comparator does not: the original penalty was not recoverable from the results, so the committed script uses `RidgeCV` over a log-spaced alpha grid, which reproduces 31 of 160 ridge rows exactly with a mean absolute difference of 0.0022 and a maximum of 0.027 on Spearman. Pool-level summary means move by at most 0.0011, so no figure quoted above changes. The ridge column is a secondary comparator and no claim rests on it alone.

2.5 What the pathogen re-point actually measured — added 19 October

This subsection moves the simulated campaign off the bacterial test protein and onto SARS-CoV-2 Mpro and two influenza haemagglutinin datasets. One earlier finding survived the move; the headline one did not, for a reason that is arithmetic rather than biological. Of the technical subsections, this is the one that constrains what may be claimed.

Action 2 below is complete, and the outcome contradicts the expected result. 240 paired replays (3 landscapes × 4 policies × 10 disjoint pools × 2 delay conditions) now exist under `runs/targets/`, summarised in `runs/summary/target_metrics.csv`.

The selected-versus-returned finding replicates. At a three-round delay the retrospective metric overstates the realtime one by 1.12–1.38× across all three landscapes and all four policies, against roughly 1.4× on GB1, and by exactly 1.00× at zero delay as it must. This is substrate-independent and endpoint-independent. It is the one quantitative result from the pilot that survived the move to pathogen targets.

The 8× policy gain does not replicate, and the cause is a metric artefact that should have been identified earlier. Random top-fraction recall is simply the labelled fraction of the pool, which the data confirms to two decimal places:

POOL	LABELLED FRACTION	PREDICTED RANDOM RECALL	OBSERVED RANDOM RECALL	HEADROOM (1/FRACTION)
GB1, 4,096 candidates	128/4096 = 0.031	0.031	0.032	32×
pathogen, 512 candidates	128/512 = 0.250	0.250	0.223	4.0×

So "8.7× better than random" is bounded above by 32× on GB1 and by 4× on a 512-candidate pool. **The multiple-over-random is not comparable across pool geometries**, and the GB1 headline figure must never be quoted without its budget ratio. That is a second measurement-design finding of the same kind as §2.2(b), and it partly undercuts §2.2(c).

Normalising as `(policy − random) / (1 − random)` removes the bound and the gain is still smaller on pathogen landscapes — 0.25 on GB1 against 0.09 for greedy on Mpro — so this is not purely an artifact. Paired per-pool differences at 128 labels, zero delay:

LANDSCAPE	GREEDY VS RANDOM	UCB VS RANDOM	DIVERSE VS RANDOM
Mpro	+0.065 ± 0.152 (7/10 pools)	+0.038 ± 0.120 (5/10)	−0.027 ± 0.065 (3/10)
HA H3	+0.104 ± 0.085 (8/10)	+0.050 ± 0.093 (6/10)	−0.031 ± 0.065 (1/10)
HA H1	−0.004 ± 0.118 (6/10)	+0.046 ± 0.107 (7/10)	+0.019 ± 0.084 (3/10)

Verdict. At a 25% labelled fraction, acquisition policy delivers a small and inconsistent gain on pathogen landscapes: best on HA H3, at chance for greedy on HA H1, and with pool-to-pool spread larger than the mean difference everywhere. Diversity-seeking is actively harmful on two of three. The likely cause is representational: a single-substitution scan encoded as (position, residue) over 303–566 positions gives the forest far less to generalise from than GB1's four co-varying sites, where every candidate shares positions with every other.

Two consequences for GATE-06. First, the geometry is a forced trade-off at 5,725 rows — many small pools compress the headroom, one large pool removes the pool-to-pool spread — and the honest comparison reports recall at a matched *budget fraction*, not at a matched absolute budget. Second, **do not carry the 8× figure into the paper.** The defensible claims are the two measurement-design findings, not the magnitude.

3. Next targets

3.1 The constraint that should drive selection

RaPID-style mRNA display requires a **purified, soluble, immobilisable target** — typically biotinylated and captured on streptavidin beads, surviving several iterative selection rounds. The library ($\sim 10^{12}$–10^{13}) is generated by flexizyme-charged initiator incorporation with spontaneous thioether macrocyclisation, so the chemistry is largely target-independent; **the reagent is the binding constraint, not the library**.

Separately, macrocyclic peptides are generally poor at crossing membranes — which is precisely why the reading list carries PeptideCLM (Feller & Wilke 2025) and the 13-method permeability benchmark (PMC12392647).

This produces a tension that should be stated rather than finessed:

> The best-behaved viral reagents are **soluble intracellular enzymes** (proteases, helicases, methyltransferases), which carry a permeability liability. The **extracellular glycoproteins** need no permeability story but are harder reagents and sit under strong positive immune selection, which undermines conservation-based family claims.

Choosing a protease and *owning* the permeability problem is more defensible than choosing a glycoprotein and quietly inheriting a conservation problem — because permeability is predictable in silico with tools already in the bibliography, whereas conservation of an immune-evasion surface cannot be engineered around.

3.2 Selection rubric

COSCI-07's acceptance condition requires documented selection criteria. Proposed rubric — these seven axes are also the backbone of the target dossier in core deliverable 2:

AXIS	WHAT IT ASKS	WHY IT GATES
Reagent tractability	Can SynX obtain soluble, correctly-folded, immobilisable protein that survives multiple selection rounds?	Hard gate. An untractable reagent ends the campaign regardless of biology.
Essentiality	Does loss of function block replication?	A binder to a dispensable protein cannot become a therapeutic.
Conservation	Is the epitope conserved across the family and under what selective pressure?	Gates every family-wide and prototype-pathogen claim.
Accessibility	Intracellular or extracellular; is the epitope solvent-exposed in the relevant conformation?	Sets whether a permeability story is required.
Assay availability	Is there a functional orthogonal assay beyond enrichment?	Without one, a single round can only substantiate enrichment.
Public label availability	Is there a DMS or binding landscape for method development?	Determines whether transfer tests are runnable now or blocked on SynX.
Precedent	Prior macrocycle or peptide campaigns against this target?	Supplies positive controls and de-risks failure interpretation.

3.3 Family 1 — Coronaviridae (prototype SARS-CoV-2)

Mpro / 3CLpro (nsp5) — primary recommendation

A cysteine protease, active only as a homodimer, with a Cys145–His41 catalytic dyad. It cleaves the viral polyprotein at eleven sites with an unusual Gln↓(Ser/Ala/Gly) specificity that has no close human protease counterpart, which helps selectivity.

AXIS	ASSESSMENT
Reagent tractability	**Strong.** Routine soluble *E. coli* expression, well-behaved, inexpensive.
Essentiality	**Strong.** Polyprotein processing is obligatory for replication.
Conservation	**Strong.** Active site highly conserved across sarbecoviruses; workable across betacoronaviruses.
Accessibility	**Weak.** Cytosolic. Requires a permeability story.
Assay availability	**Strong.** Standard FRET peptide-cleavage assay for functional follow-up.
Public labels	**Yes** — `R1AB_SARS2_Flynn_2022`, 5,725 mutants, FRET + growth readout, already downloaded.
Precedent	Published macrocyclic peptide campaigns exist, including RaPID-derived work. **Verify specific citations before treating any as a positive control.**

The cytosolic liability may be an asset for this project. It forces a genuinely multi-objective decision — affinity against predicted permeability — which is a harder and more interesting problem for a co-scientist than single-score ranking, and it activates an entire section of the reading list that would otherwise go unused.

PLpro (nsp3 papain-like protease) — strong secondary

Cleaves the nsp1/2/3 boundaries and additionally strips ubiquitin and ISG15 from host proteins, so inhibition is simultaneously antiviral and immune-restorative. Soluble and tractable; a zinc-binding domain requires attention in construct design. Conserved but less so than Mpro across genera. No public DMS. Cytosolic.

nsp10/nsp16 interface — the interesting protein–protein option

nsp16 is a 2'-O-methyltransferase that caps viral RNA to evade innate immune sensing, and it is **catalytically dependent on nsp10 as an obligate cofactor**. The nsp10–nsp16 interface is therefore a protein–protein interaction that macrocycles are constitutionally better suited to than small molecules are. Conserved, essential for immune evasion. Harder reagent (two-protein complex), no public DMS, cytosolic.

This merits attention because "macrocycles outperform small molecules at flat PPI interfaces" is the strongest mechanistic argument for the modality, and a PPI target lets the project make that argument rather than competing with small molecules on an enzyme active site.

nsp13 helicase

Superfamily-1 helicase with 5'→3' polarity, NTPase activity, zinc-binding domain. Highly conserved, essential. Reagent is moderately tractable. Cytosolic, no public DMS.

Spike RBD — best data, worst conservation argument

`SPIKE_SARS2_Starr_2020_binding` (3,802 variants) and `_expression` (3,798) are already on disk and are among the cleanest binding landscapes in existence. Extracellular, so no permeability story needed, and the reagent is well established.

But the RBD is an immune-evasion surface under intense positive selection. It is therefore the **wrong choice for any conservation or family-transfer claim**, despite the data quality. Use it for method development; do not build the family narrative on it.

Spike S2 stem-helix / fusion peptide — the conserved extracellular option

Far better conserved across betacoronaviruses than the RBD, and the target of known broadly-neutralising antibodies, which establishes that the epitope is accessible and druggable. Harder reagent — needs a stabilised construct or peptide presentation rather than whole prefusion spike — and no public DMS. This is the right answer if the project needs a conserved *extracellular* coronavirus target.

3.4 Family 2 — two options, of which the more obvious is not the better experiment

Henipavirus (Nipah) — better narrative

- **G attachment glycoprotein.** Tetrameric, with a six-bladed β-propeller head that binds ephrin-B2 and ephrin-B3. Genuinely extracellular, so no permeability story. Receptor-binding site is well mapped. Reagent is obtainable as a soluble head domain.
- **F fusion glycoprotein.** Class I trimeric fusion protein requiring cathepsin L cleavage for activation. The prefusion conformation is metastable and needs stabilisation to be a usable display reagent — a genuine reagent risk.

Strengths: WHO R&D Blueprint priority pathogen; a real conservation ladder for transfer testing (NiV-Bangladesh → NiV-Malaysia → Hendra → Cedar → Langya); matches the "extracellular peptides" framing from the Oct 5 discussion. Weakness: **no public DMS**, so the labeled molecular-transfer tests stay blocked on SynX data.

Influenza A — better experiment

- **HA stem (HA2 helix A region).** The conserved epitope recognised by group-1-specific and group-2-specific broadly-neutralising antibodies. Extracellular, well conserved within a group, strong peptide-binder precedent.
- **NA catalytic site.** Sialidase active site, highly conserved framework and catalytic residues, extracellular, validated as a drug target.
- **PA–PB1 interface.** The PA C-terminal domain binds an N-terminal PB1 peptide — a well-characterised PPI with published peptide inhibitors. Intracellular.

Decisive advantage: **four influenza landscapes are already downloaded** — HA for H1N1 (Doud 2016, 10,715 mutants) and H3N2 (Lee 2018, 10,754), NA (Jiang 2016), and polymerase subunits. The H1→H3 transfer test is runnable this week on data already on disk, which is exactly the grouped-holdout shift test the brief specifies.

Recommendation

Coronaviridae (Mpro) + Orthomyxoviridae (HA stem / NA).

Both families have public labeled data, which unblocks the molecular transfer tests immediately rather than making them dependent on a collaborator. Together they also span the tension in §3.1 deliberately — one soluble intracellular enzyme with a permeability problem, one conserved extracellular epitope without one — which makes the pair a designed contrast rather than two arbitrary picks.

Retain **Nipah as the stretch robustness family**: better story, no labels.

3.5 Pathogen landscapes already on disk

DMS ID	TARGET	ASSAY	MUTANTS
`R1AB_SARS2_Flynn_2022`	SARS-CoV-2 Mpro	FRET, growth	5,725
`SPIKE_SARS2_Starr_2020_binding`	SARS-CoV-2 Spike RBD	ACE2 binding	3,802
`SPIKE_SARS2_Starr_2020_expression`	SARS-CoV-2 Spike RBD	ACE2 binding	3,798
`A0A2Z5U3Z0_9INFA_Doud_2016`	Influenza HA (H1N1)	Viral replication	10,715
`C6KNH7_9INFA_Lee_2018`	Influenza HA (H3N2)	Viral replication	10,754
`NRAM_I33A0_Jiang_2016`	Influenza neuraminidase	—	298
`A4D664_9INFA_Soh_2019`	Influenza PB2	Viral replication (avian cells)	14,421
`RDRP_I33A0_Li_2023`	Influenza PB1	Viral replication	12,003
`POLG_DEN26_Suphatrakul_2023`	Dengue NS5	Viral replication	16,897
`A0A140D2T1_ZIKV_Sourisseau_2019`	Zika envelope	Viral replication	9,576

Engineering note: Spike RBD has 3,802 variants and Mpro 5,725, both at or below the current `pool_size: 4096`. The twenty-disjoint-pool design needs rework for these — either fewer, smaller pools, or resampling with documented overlap. Do not silently reuse candidates across "disjoint" pools.

3.6 Other families, for completeness

- **Flaviviridae NS2B–NS3 protease** (dengue, Zika). Soluble, standard fluorogenic assay, requires the NS2B cofactor peptide for activity. Among the most tractable viral protease reagents available, and dengue NS5 and Zika envelope landscapes are already on disk. A credible third family if influenza is rejected.
- **Filoviridae GP** (Ebola, Marburg). The NPC1-binding site is only exposed after cathepsin cleavage, which makes the conformational state of the reagent the central design question. High-value, harder.
- **Arenaviridae GPC** (Lassa). Prefusion trimer, notoriously difficult to stabilise. Not recommended as a first campaign.
- **Bunyavirales Gn/Gc** (Rift Valley fever, CCHF). Both WHO priorities; reagents less established.

3.7 The ambitious alternative worth 10 minutes at the gate

The reading list already contains PMC12464979 — macrocyclic binders, covalent modifiers and degraders of a **structured RNA**, discovered by mRNA display.

The SARS-CoV-2 **frameshift stimulatory element** is a pseudoknot between ORF1a and ORF1b that sets the −1 programmed ribosomal frameshift ratio. It is highly conserved (structured RNA elements are constrained by base-pairing, not just by sequence) and functionally essential, since the virus requires a specific ratio of ORF1a to ORF1b products.

Targeting it would be the most novel element available to the project, has direct precedent in the project's own bibliography, and sidesteps the conservation problem entirely because RNA secondary structure is under far stronger constraint than a surface glycoprotein epitope. Against it: non-standard reagent and assay, a less certain functional readout, and no public landscape. Worth putting to Askar even if declined.

3.8 Recommendation for the single display round (LAB-01, due Oct 23)

Mpro. Cheapest and best-behaved reagent, strongest conservation case, an orthogonal functional assay available if the schedule allows, and interpretable failure. Report enrichment as enrichment — one round does not establish affinity, functional inhibition or antiviral activity.

4. Next steps and co-scientist architecture

4.1 The gap

The agent's current decision state is:

```
observed_score_summary, candidate_summary[{candidate: <int>, prediction,
uncertainty_heuristic, diversity}], remaining_label_budget, pending_labels,
feedback_delay_rounds, decision_round
```

with an action space of `{greedy, explore, diverse}`. Candidate identities are deliberately opaque integers.

So what exists is a **bandit-strategy selector**. It reasons competently about budget and uncertainty, but it contains no biology, and it is several abstraction layers from the proposal's Part 1: pathogen intake from genomic sequence → evidence retrieval → structure prediction → target and epitope ranking → library design.

Closing that gap means the decision state becomes the **target dossier schema** — the seven axes in §3.2 — and the action space becomes what the brief already specifies: retrieve missing evidence, request a structure calculation, revisit a target ranking, invoke a design tool, recommend an experiment. The budgeted-acquisition machinery already built becomes the resource accounting underneath that policy.

4.2 Architecture brainstorm: treating this as a data problem, not a prompting problem

This section considers how a model might be *trained* to make these decisions well, rather than instructed more effectively. It is the least settled part of the document. The practical content lies in the first two items: public measurement datasets can be converted into an effectively unlimited supply of automatically gradeable practice problems, which is inexpensive and appears to be the genuinely novel element; and an expensive model can then serve as a teacher for a cheaper one, which is what would render a search over many candidate next actions affordable. The remaining items refine the training and evaluation plan. Only the first is a ten-week deliverable, as §4.3 states.

The five capabilities raised in review map onto the existing Week 8–10 harness more closely than anticipated. Each is set out below with its interpretation here and an assessment of how much is already planned as against genuinely new.

(a) Scale synthetic episodes with containerised verification

Pattern: generate synthetic tasks at volume, verify each one automatically in a container, keep only what passes.

The scientific analogue is strong. Public DMS landscapes are **auto-gradeable decision environments**: the answer is already known for every variant, so a proposed acquisition, ranking or stop decision can be scored deterministically with no human and no wet lab. Each of the ten landscapes in §3.5 generates an effectively unbounded number of episodes by varying pool composition, budget, delay, initial evidence, corrupted evidence and conflicting evidence. The containerised oracle holds the hidden outcomes, exposes only a tool API, and returns a verified score — the same isolation boundary already enforced in `pick_batch`, just packaged.

Already planned: the synthetic world generator with known latent mechanisms and an independently validated oracle, plus the five stress conditions (clean, confounding, mechanism shift, evidence corruption, longer horizon).

Genuinely new: generating episodes from the *molecular* landscapes rather than only the transcriptomic worlds, which would let the applied SynX case and the forecasting testbed share one episode format.

Scope qualification: the pattern scales to trillions of tokens in a pretraining context. This project budgets ~12.5M training tokens total. The verification architecture is the transferable part; the scale is not, and claiming otherwise in a paper would be indefensible.

(b) Distil a flagship model into a fast policy

Pattern: generate reasoning traces with an expensive model, train a cheap model on them.

This is the missing piece that makes the rest work. The current agent makes one call per decision round at ~2.8 s. A co-scientist that *searches* over actions — evaluating several candidate next-actions before committing — needs hundreds to thousands of rollouts per episode. That is unaffordable at flagship prices and latency, and it is the real reason the current design collapsed into a three-option strategy selector rather than an action search.

The route: run a flagship model on verified episodes from (a) with full evidence retrieval and extended rationale, keep only trajectories that the container scored well, and use those as the training curriculum for the small trainable bases. The existing two curricula — conclusion/evidence reasoning, and grounded tool use/planning — are exactly the right two buckets, and distilled flagship traces are a far better source for them than hand-written data.

Status: not in the plan. This is the most valuable addition and it slots into the existing curriculum slots without changing the experiment matrix.

(c) Mid-training data mixtures to raise the RL ceiling

Pattern: the mixture determines the headroom that later RL can exploit.

The question for this project: what mixture of protein sequence data, structural data, assay protocols, primary literature and decision traces does a base model need before decision-quality RL has anything to climb? A model that has never seen an assay protocol cannot learn to sequence experiments, however good

the reward.

Already planned, with the variable already in place: the matrix has 2 curricula × 3 models × 2 seeds, and the uplift metric `U(BT, BI) = C(post-trained, BI) − C(base, BI)` with conclusion, grounding and process losses reported separately. That *is* a mixture ablation with a ceiling measure attached.

The sharpening: treat curriculum composition as the designed variable rather than a fixed pair of labels, and report uplift as a function of mixture rather than only of budget. Decomposed losses make this readable — a mixture that improves grounding while flattening conclusion quality is a result, not noise.

(d) Long-context data recipes

Pattern: deliberately engineer how long inputs are composed, not just how long they are.

This is a structural property of the problem, not an add-on. A target dossier accumulates monotonically within an episode: genome, MSA, structure predictions, conservation analysis, literature, prior assay rounds, and the rationale for every earlier decision. By the final decision of an episode the evidence state is the largest object in the system, and **what gets evicted from it is itself a decision the policy should make**.

Concrete recipe questions: does the dossier stay as raw evidence or as a running summary; is prior rationale retained verbatim, summarised, or dropped; when evidence conflicts, does the newer entry replace the older or sit alongside it with both timestamps. The brief's requirement that the decision state record the evidence available at that moment means conflicting evidence must coexist rather than being overwritten — which has a direct and under-appreciated context cost.

Partially planned: the inference budget (4,096 generated tokens, 8 tool calls) with a control arm at 8,192 and 16 calls, plus "longer horizon" as an explicit stress condition, which separates post-training gains from extra inference. The eviction policy itself is not yet specified.

(e) Robust, diverse evaluation of mid-training checkpoints

Pattern: without a cheap, reliable, diverse eval, neither mixture ablation nor RL can be steered.

This is already the best-specified part of the plan, and it is worth recognising as such: 51 model states (48 trained checkpoints at 16K / 64K / 256K / 1M tokens, plus 3 zero-shot bases) × 5 stress conditions × 10 paired cases × 2 rollout seeds = 5,100 episodes, plus 600 inference-control episodes. Cases are allocated across four development world families, three unseen mechanism families and three audited real studies, with holdouts grouped by entire model and curriculum groups rather than by budget point or rollout seed.

The addition: the verified molecular episodes from (a) become additional eval conditions, giving the checkpoint suite a domain where ground truth is exact rather than expert-assessed. That is a useful complement to rubric-scored episodes, whose inter-rater reliability is itself a project risk.

4.3 The reframe this suggests

Taken together, the five capabilities point at a sharper contribution than "an agent that helps scientists decide."

> The bottleneck in scientific-decision agents is not the policy architecture. It is the absence of **verifiable decision environments** at the volume needed to train and evaluate one. Public DMS landscapes are auto-gradeable decision environments that already exist, in quantity, and are not being used that way.

The project is unusually well placed to argue this: it already has the harness, the uplift metric, the checkpoint grid, and — as of §3.5 — ten landscapes on disk. The pipeline that turns them into verified episodes is the missing component, and it is smaller than the harness that would consume it.

This also rescues the Oct 12 work: the GB1 pilot is the first instance of exactly this pattern — a verified, auto-graded decision environment built from a public landscape with enforced label isolation. It was built as a biology pilot, which is why it looks weak. As infrastructure it is the prototype for (a).

Principal risk: a mid-training data programme is not a ten-week deliverable. The minimum viable version is the episode generator plus containerised oracle over two or three landscapes, feeding the eval conditions in (e). Distillation (b) and mixture ablation (c) are W8–W9 at the earliest and only if access and spend gates clear.

4.4 Ordered actions

1. **Oct 19 — COSCI-07.** Put Mpro + influenza to Askar with the §3.2 rubric as the documented selection criteria and §3.5 as the availability argument. Gates MOL-06, LAB-01 and GATE-06. **Packets now exist** — `hpc_pilots/targets.py` encodes 11 targets across three families against the seven axes, with each claim's provenance and three claims flagged as needing a citation. The decision itself is still Askar's.
2. ~~Re-point the binding harness~~ — **done.** Mpro, HA H1 and HA H3 are staged as decision environments in `hpc_pilots/configs/targets.json`. The pool-size rework was real: 20 × 4,096 disjoint pools is impossible at 5,725 rows, so pools are 512 candidates and the geometry shrinks rather than reusing candidates — 10 disjoint pools for Mpro, 19 for HA H1, 20 for HA H3. Top fraction moves to 5% because 1% of 512 is about five members; those figures must not be compared with the GB1 1% figures.
3. **Retire the structural arm** as a scientific component, per §2.3. Keep the cost-accounting and evidence-isolation code.
4. ~~Write the dossier schema before the policy that consumes it~~ — **done.** `hpc_pilots/dossier.py` holds the seven axes as the decision state with per-axis citations, timestamps, retained conflicts and hard-gate blocking, and the six-action space replacing `{greedy, explore, diverse}`. COSCI-06 now depends on it in the plan, so the policy cannot be built first.
5. ~~Specify the episode format once~~ — **done for the molecular side.** `hpc_pilots/emergency.py` defines the episode: timed evidence release, a deadline, an action budget, and scoring on updating under

relevant evidence versus holding under irrelevant evidence. The transcriptomic episodes in SAFE-04 must adopt this format rather than forking it.

6. ~~Restore the calibration scripts~~ — **done.** `hpc_pilots/calibration.py` now regenerates both studies; see §2.4 for the one column that does not reproduce exactly.
7. **Pull Rogers, Passioura & Suga (2018)** nonproteinogenic DMS (PNAS 115(43), 10959–10964) — a deep mutational scan of linear *and cyclic* peptides including non-canonical residues, the only labeled macrocycle dataset with ncAAs in the bibliography, and a far better substrate for low-label ranking work than GB1.
8. **Prototype the containerised oracle** over one landscape before committing to the full episode generator.

5. Open items

- **Pool geometry for the pathogen landscapes is unresolved.** §2.5 shows the multiple-over-random is bounded by the labelled fraction, so the current 512-candidate pools cannot reproduce a GB1-scale effect even in principle. Decide at GATE-06 between many small pools with compressed headroom and few large pools with no spread, and report matched budget fractions either way.
- **The LLM policy has no non-smoke result anywhere.** `runs/summary/binding_metrics.csv` contains only the four heuristic baselines; the only agent run on disk is a Haiku smoke run. The emergency episodes have been run with the two deterministic policies only, and those are stable under irrelevant evidence by construction, so the stability metric has not yet been tested against anything that could fail it. COSCI-06 is where that happens.
- The non-canonical building-block registry carries literature block *classes* only. Every chemical identifier, codon assignment, Rosetta parameter and rotamer library is null pending the MOL-01 export, so no macrocycle has been represented concretely, let alone measured.
- The ridge comparator in §2.4 does not reproduce bit-exactly (mean absolute difference 0.0022); the forest and calibration columns do.
- No prototype family, target, assay or chemical alphabet selected. COSCI-07 is the blocker.
- The entire physics-based ncAA track remains untouched — Flex ddG, FakeRotLib, MakeRotLib, Dunbrack rotamer libraries, the RosettaCommons ncAA interface protocol. Nothing in the repository approaches it.
- Macrocyclic-peptide precedent against Mpro is asserted from memory in §3.3 and must be verified before it is used as a positive-control argument.
- GATE-06 additionally depends on the lab route and data-return terms, live trainable model access and spend cap, and the novelty review — all external to this repository.
- ATLAS data is CC BY-NC 4.0; this work is scoped to noncommercial research. Commercial use requires separate terms from the dataset owner.

6. Glossary

Every machine-learning and statistics term used above, defined without assuming a computational background. Grouped by where the terms appear rather than alphabetically.

The simulated campaign (§1, §2.2, §2.5)

Candidate / pool. A candidate is one variant whose measurement could be requested. A pool is a fixed set of candidates treated as one self-contained trial. Pools are **disjoint**, meaning no candidate appears in two of them, so each pool is an independent repeat of the same exercise.

Decision policy / acquisition policy. The rule for choosing which candidates to measure next. This is the object under study. Four were compared: **random** (pick blindly, the control), **greedy** (pick whatever the predictor rates highest), **UCB** (pick things that are either rated highly or poorly understood, so it deliberately spends some budget on reducing its own ignorance), and **diverse** (pick things chemically unlike what has already been measured).

Predictor / surrogate. A cheap statistical model that guesses a candidate's result before it is measured, fitted only on results already revealed. It stands in for the real measurement, hence "surrogate". Deliberately kept weak and simple here, so that any advantage is attributable to the choosing rule rather than to a clever model.

Random forest. A standard off-the-shelf predictor that averages many simple decision trees. **Ridge regression** is a much simpler standard predictor, included only as a sanity check: if the simple one does as well as the complicated one, the complicated one is not contributing anything.

Features / encoding. How a candidate is converted into numbers a model can take as input. Here it is minimal on purpose: only which position is mutated and to which residue. No structure, no evolutionary information, no sequence context.

Budget / batch / decision round. The campaign may request 128 measurements in total, eight at a time. Each request-and-wait cycle is one decision round.

Delayed feedback. Results arriving some rounds after they were requested, so the policy must keep choosing while earlier requests are still outstanding. The delays used (zero and three rounds) are deliberately artificial stress settings, not estimates of real assay turnaround.

Held out. Data deliberately withheld so it can be used to test whether a method generalises. A **grouped** holdout withholds an entire natural group — a whole protein, a whole study, a whole subtype — rather than a random scattering of rows, which is a much harder and more honest test.

Scoring (§2.2, §2.4, §2.5)

Top-fraction recall. Of all the genuinely best candidates in a pool — say the best 5% — what share did the campaign actually find? This is the primary score.

Selected versus returned. Two ways of computing that score. **Selected** counts everything the policy chose, including requests whose results had not yet come back when it was choosing. **Returned** counts only what had actually arrived. Selected is the hindsight-flattered version; returned is the fair one. The gap between them is the finding in §2.2(b) and §2.5.

Enrichment ratio. The degree to which the recovered set is richer in strong candidates than the library it was drawn from. A value of 1.0 indicates no improvement over the starting library. This is the form of number an mRNA display round actually produces, which is why it was added.

Normalised gain. A policy's improvement expressed as a share of the improvement that was available to it, rather than as a raw multiple. Used in §2.5 because raw multiples are not comparable between pools of different sizes.

Spearman correlation. Agreement in *ranking*, from 0 (no relationship) to 1 (identical order). It ignores whether the predicted values are the right size and asks only whether things are put in the right order.

Calibration, overconfidence ratio, 1σ coverage. Calibration is whether a predictor's stated uncertainty matches its real error. The overconfidence ratio is how many times larger the real error is than the predictor claims; 1.0 would be honest, 63 is not. 1σ coverage is the share of cases falling inside the predictor's own stated error bar, which should be about 0.68 if those bars are trustworthy.

Paired comparison, ± spread, "7/10 pools". All policies start each pool from the same eight initial measurements, so they can be compared pool by pool rather than only on averages. The ± figure is the spread of that per-pool difference, and "7/10 pools" is how many pools the method actually won. When the spread is larger than the average difference and the method wins only 7 of 10 pools, the effect is weak — which is the honest reading of §2.5.

The decision system (§4.1, §4.2)

Bandit. A family of algorithms that decide only which option to try next and contain no understanding of the subject matter. Calling the earlier system "a bandit-strategy selector" means it chose between abstract strategies without knowing it was doing biology. Replacing that was the point of the 19 October work.

LLM (large language model). A model trained on text that can follow instructions and produce reasoning in prose; the component here that reads a dossier and proposes the next action.

Decision state / action space. The decision state is everything the system can see when it chooses; the action space is the set of moves available. The state is now the seven-axis target dossier, and the moves are: retrieve evidence, request a structure calculation, re-rank targets, propose a library, recommend the

experiment, stop.

Episode. One simulated decision scenario from start to finish: a set of candidate targets, evidence arriving on a timetable, a deadline, and a budget of actions.

Tokens / context. Tokens are the units of text a model reads and writes, used as the unit of cost. Context is how much text it can hold in working memory at once — which matters because a dossier grows as evidence accumulates, and something eventually has to be dropped.

Checkpoint. A saved snapshot of a model partway through training, so that performance can be measured as a function of how much training has been done.

Distillation. Running an expensive, capable model to generate worked examples, then training a cheap model to imitate them. The motivation here is cost: searching over many possible next actions requires hundreds of model calls per decision, which is unaffordable at expensive-model prices.

Mid-training data mixture. The proportions of different kinds of text a model is trained on before any task-specific tuning. The claim it raises is that a model which has never seen an assay protocol cannot learn to sequence experiments, however it is rewarded afterwards.

Reinforcement learning (RL). Training by reward rather than by imitation: the model acts, is scored, and is adjusted toward whatever scored well. Needs a reliable scorer, which is why the auto-gradeable datasets in §4.2(a) matter.

Uplift. How much better a trained model is than the same model untrained, measured on the same problems. The project's headline measurement.

Containerised oracle. A sealed program that holds the hidden answers and will only confirm or score what it is asked, so the system being tested cannot read them. The software equivalent of keeping the answer key in another room.